Tut2

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Part I
UFD and prime factorization.

1 Definitions

- **Definition:** A ring R is a PID if

1. R has no zero divisors (integral domain)
2. Every ideal I is of the form (r) for some $r \in R$.

- We call (1) the trivial ideal.

1.0.1 Proposition

- $\mathbb{Z}$, $k[X]$ are PID's.

1.0.2 Proof

- Given $I \subset \mathbb{Z}$, consider $i \in I$ with $|i|$ minimal. We claim $I = (i)$. Indeed, any
$$n = qi + r$$
with $|r| < i$ (Draw picture). Thus if $n \in I$, then $n - qi \in I$, so r has to be zero.

- Given $I \subset k[X]$, consider $i \in I$ with $deg(i)$ minimal....

1.0.3 Example

- Given $i, j \in \mathbb{Z}$, then (i, j) is principal, and the proof gives an algorithm to find $d = gcd(i, j)$, that's the Euclidean algorithm.

1.0.4 Lemma

- If R is a PID and p irreducible, then (p) is a prime ideal- i.e., if $p|ab$, then it divides either a or b.
- **Proof:** Suppose that $p = ab$ and $p \nmid a$, we want to express that p, a are coprime. Then $(g) = (p, a) \subsetneq (p)$, so $g \mid p$, but p is irreducible, and $p \nmid a$, so g must be a unit. That's coprimeness! Thus $\exists \alpha, \beta$ with

  $$\alpha a + \beta p = 1$$

  multiplying by b we obtain

  $$\alpha ba + \beta bp = \alpha p + \beta bp = b$$

  and so $b \in (p)$.
- **Philosophy of the proof:** You see that the correct definition of coprime is a **constructive one.** Good definitions are not negations "p doesn't divide a", but affirmations of data ($\alpha a + \beta p = 1$) that may prohibit other things.

2 Theory- PID is UFD

2.0.1 Combinatorial lemma

- If T is a tree of finite degree at each vertex that doesn't contain any infinite path, then T is a finite tree.
- **Proof sketch:** Suppose by contradiction T has infinitely many vertices. Then by induction, we can find a compatible family of paths P_k of length k which lead to a tree with infinitely many vertices. Then $(P_1, P_2, ...)$ is an infinite path.

2.0.2 Proposition

Any ring R that is a PID is UFD

Proof:

- First, we show any $a \in R$ has a factorization to irreducibles. If a is not irreducible, then $a = bc$. Note $(b), (c) \supset (a)$. If $(b) = (a)$, then

  $$a = ab'c \to 1 = b'c \to c$$

  is a unit. Thus we may assume $(b) \supsetneq (a)$. We can draw a tree corresponding to the factorizations. By the lemma there is an infinite path. We find

  $$(a_0) \subsetneq (a_1) \subsetneq (a_2) \subsetneq ....$$

  the union I is an ideal, so of the form (i). But then $i \in a_n$ for some n, contradicting $(a_{n+1}) \neq (a_n)$. Thus the tree has finitely many vertices, corresponding to a finite factorization to irreducibles.

- Second, we show any factorization is unique. Suppose that

$$p_1...p_m = q_1...q_n$$

for irreducibles p_i, q_j(repetitions allowed). We wish to show that up to units we can match both sides (imagine

$$2 \cdot 3 = -2 \cdot -3$$

).

- Thus $p_1|q_1....q_n$, so $p_1|q_i$ for some i, but then $p_1 = q_i$ up to a unit, and we may cancel them out and continue.

2.0.3 Proposition- aposteriori

- Let R be a PID, then a, b don't share a prime divisor iff they are coprime (i.e., $\alpha a + \beta b = 1$).
- **Proof:**
- If $\alpha a + \beta b$, then any $p|a, b$ is a unit
- If a, b don't share a prime divisor, then $(a, b) = (i)$ since R is a PID, and i must be a unit otherwise a, b would share a divisor.

3 Solve exercises $7, 8, 9$

Part II
Review of algebraic integers

- **Definition:** Let $i : R \to S$ be a ring extension (in our case, $\mathbb{Z} \hookrightarrow K$ for a number field K), then $s \in S$ is algebraic over R if $\exists r_0, ..., r_{n-1}$with

$$s^n + i(r_{n-1})s^{n-1} + ... + i(r_0) = 0$$

- When $R \hookrightarrow S$ we can ignore i and abuse notation as $R \subset S$.

3.0.1 Definition

- $c_n x^n + ... + c_0 \in \mathbb{Z}[x]$ is primitive if $gcd(c_i) = 1$.
- Equivalently, f is primitive if for all primes p, $\bar{f} = 0$ (where $\bar{f}$ is the reduction mod p).

3.0.2 Gauss's lemma

- If f, g are primitive, so is $h = f \cdot g$.
- **Proof:** Suppose by contradiction $\bar{h} = 0$, then $\bar{f} \cdot \bar{g} = 0$, but $\mathbb{F}_p[x]$ is an integral domain.

3.0.3 Remark

- Gauss's lemma holds when replacing $\mathbb{Z}$ via any PID R, redefining primitive appropriately.

3.0.4 Proposition

- For R a PID, and $R \hookrightarrow Frac(R) \hookrightarrow S$ (think of $\mathbb{Z} \hookrightarrow K$ a number field), $s \in S$ is algebraic iff its minimal polynomial $p(x) \in Frac(R)[x]$, normalized to be monic, has coefficients in R.
- **Proof:** If the minimal polynomial $p(x)$ of s satisfies that, then s is algebraic over R. Conversely, if $\exists q(x)$ with $q(s) = 0$ and q monic with coefficients in R, then
$$q(x) = p(x)h(x)$$
for $h(x) \in Frac(R)[x]$. We claim that $p(x), h(x) \in R[x]$, which implies $p(x)$ is monic. Indeed by taking the lcm of the reduced denominators of the coefficients, we may renormalize $\tilde{p}(x) = p(x) \cdot l, \tilde{h}(x) \cdot l'$, $l, l' \in R$ s.t $\tilde{p(x)}, \tilde{h}(x)$ are primitive. Thus
$$l \cdot l' \cdot q(x) = \tilde{p} \cdot \tilde{h}$$
. Since the right hand side is primitive, $l \cdot l'$ is a unit, so l, l' are units, and thus $p(x), h(x) \in R[x]$.

3.0.5 Corollary

- $\mathbb{Z} \subset \mathbb{Q}$ are precisely the algebraic integers in $\mathbb{Q}$.

Part III
The Rosetta stone of mathematics

Why are we taking the algebraic closure of $\mathbb{Z} \subset K$, what does it mean? There is a satisfying answer, but it requires the patience and difficult mind shift to algebraic geometry.

3.0.6 Slogan

There is a **powerful** analogy between the rings

$$\mathbb{Z}, \mathbb{F}_p[x], \mathbb{C}[x]$$

In fact, the way modern number theory works is via proving results on $\mathbb{C}[x]$, and mimicking them for $\mathbb{Z}$.

First, let's advertise the relationship.

3.1 Examples of comparisons

1. Both are PID's.
2. Both are Dedekind domains.
3. The number of primes up to n in $\mathbb{Z}$ is $\sim n/ln(n)$. The number of irreducible polynomials of degree d ("up to p^{d+1}") in $\mathbb{F}_p[x]$ is $\sim p^{d+1}/d+1 = p^{d+1}/log_p(p^{d+1})$.
4. Both satisfy quadratic reciprocity.

3.1.1 Examples of definitions\theorems algebraic geometry clarifies

1. Integrally closed $\to$ normal curve.
2. Ramification (it's a geometric word!)
3. Inertia group (it's a geometric word!)
4. Galois extension $\to$ etale cover.
5. Dedekind domain $\to$ smooth curve.
6. Minkowski's theorem $\to$ Riemann-Roch.
7. Finiteness of class groups $\to$ finite dimension of Jacobian of curve.

3.2 Perspective change

An ideal mathematical education would **start with** $\mathbb{F}_p[x]$ and introduce $\mathbb{Z}$ later. It's a coincidence that $\mathbb{Z}$ appears in our life as kids, and so we find it more intuitive. But after completing the shift, many people find $\mathbb{C}[x]$ more intuitive.

3.3 Why is the analogy true?

There's no very satisfying answer I know of, but it's an experimental fact. We'll just assume the analogy, and redevelop the theory on the side of $\mathbb{C}[x]$.

4 What is algebraic geometry

- Here are ways to create rings

1. Fix a topological space X and consider $C^{cont}(X, \mathbb{R})$
2. Fix a smooth manifold M and consider $C^{\infty}(M, \mathbb{R})$
3. Fix a complex manifold M and consider $C^{hol}(M, \mathbb{C})$
4. Fix a complex algebraic manifold M and consider only **algebraic** functions $C^{alg}(M, \mathbb{C})$

- Here are ways to create maps between rings

1. For a continuous map $f : X \to Y$ we get $C^{cont}(Y, \mathbb{R}) \to C^{cont}(X, \mathbb{R})$
2. For a smooth map ...
3. For a complex holomorphic map...
4. For a complex algebraic map...

What these all have in common is that "sorta functions" on a "sorta space". Since humans have great vision and geometric intuition, algebraic geometry is the completely fake, syntactic lie that any ring R (including $\mathbb{Z}$, including $\mathbb{F}_p$, including $\mathbb{C}[x]$, including $\prod_{\mathbb{N}} \mathbb{Z}$, including $\mathbb{Q}_p$) is a "sorta functions" on a "sorta space". The space corresponding to a ring R is called $Spec(R)$, it is a topological space. **Thus the functions on** $Spec(R)$ is **rigged** to be the ring R.

The definitions are **rigged** so that the geometric maps $Spec(R) \to Spec(S)$ **correspond** to pullback on functions maps $S \to R$. So all maps of rings are rigged to be geometrically intuitive.

5 Precise definitions

Sadly, algebraic geometry has a wealth of definitions that take a while to digest. We'll just give the most basic one:

5.1 *Spec*

The topological space $Spec(R)$ has points the **prime ideals** $\mathfrak{p}$ of R.

Aposteriori, if we think of $Spec(R)$ as an existing mathematical concept, we think of $\mathfrak{p} \subset R$ as the functions on $Spec(R)$ that vanish on the point $\mathfrak{p} \in Spec(R)$.

That suggests the topology, a function should vanish on a closed subset, and so for any $f \in R$, the collection $V_f = \{\mathfrak{p} | f \in \mathfrak{p}\}$ is a closed subset, and that defines the topology.

5.1.1 Examples

- For a field k, $Spec(k)$ is a point.
- $Spec(\mathbb{C}[x])$ has one generic point (0), and a point $(x-r)$ for $r \in \mathbb{C}$.
- $Spec(\mathbb{Z})$ has one generic point (0), and a point (p) for each prime $p \in \mathbb{Z}$.

5.1.2 Exercise

- Check that if $R \to S$ is a ring map, there is a continuous map $Spec(S) \to Spec(R)$.

5.2 Functions

$r \in R$ is a function on $Spec(R)$ that takes the value $\bar{r}$ on $\mathfrak{p} \in Spec(R)$ (Where $\bar{r} \in R/\mathfrak{p}$ is the reduction of r).

Thus functions take values in different rings in different points....

6 Upshot

We can draw imprecise pictures of the sorta space of a given ring, and sometimes deduce\guess theorems about the rings using our geometric intuition!

6.0.1 Elementary examples

- If $R \twoheadrightarrow S$ is a square zero extension (i.e., $I \to R \to S$ has $I^2 = 0$), and $s \in S$ is an idempotent, then for any lift $\tilde{s}$, then $3\tilde{s}^2 - 2\tilde{s}^3$ is an idempotent.

6.0.2 Clever examples

- R being integrally closed in $Frac(R)$ is related to the smoothness of $Spec(R)$. If you imagine a geometric smooth curve C, and $R = \Gamma(C)$. And a rational function $h = \frac{f}{g} \in Frac(R)$ with $h \notin R$ (so h has **poles** at a point $c \in C$), then h cannot satisfy a monic polynomial over R. Indeed if
$$h^n + a_{n-1}h^{n-1} + ... + a_0 = 0$$
then at $c \in C$, the left hand side poles at least n times.